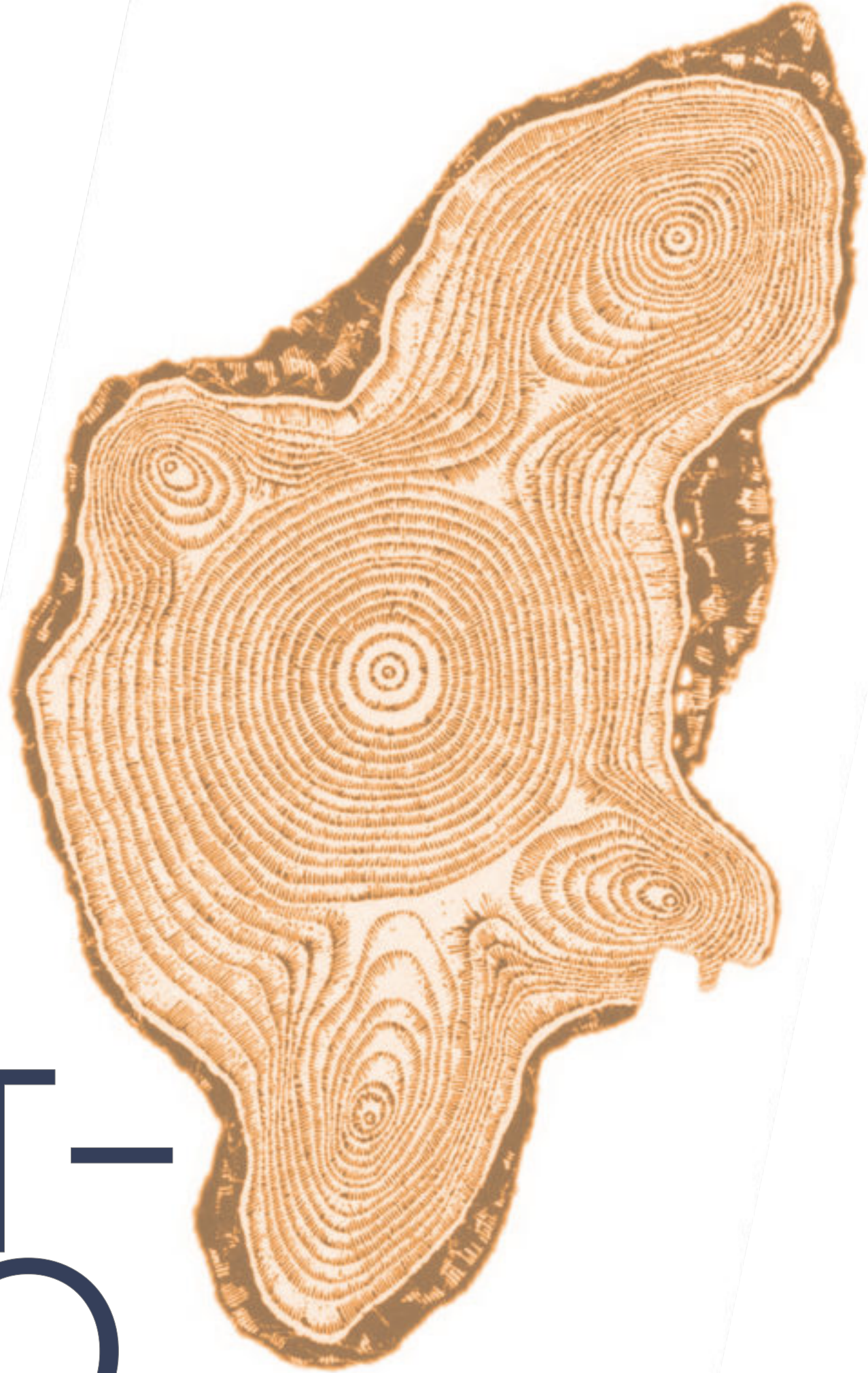


AD2 | RETHINKING LOGISTICS

# PORT- FOLIO



## STUDIO BRIEF

The studio investigates logistics architecture as a dynamic part of the urban and ecological landscape rather than treating it solely as a technical or industrial necessity. Working alongside professional collaborators including Ramboll, Henning Larsen, and Panattoni, the research addresses contemporary challenges connected to the development and operation of logistics environments.

The approach combines research-driven analysis with experimental design strategies, using quantifiable indicators such as carbon impact, energy performance, and biodiversity to guide architectural decisions. By questioning conventional warehouse typologies, the studio explores new possibilities for logistics spaces that are flexible, environmentally responsive, and better integrated into broader urban systems.



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# Redefining Logistics Architecture as Urban and Ecological Infrastructure



Logistics is no longer a peripheral industrial necessity; it is the heartbeat of the modern world. This studio investigates the shift from the 'Big Box' warehouse to a dynamic architectural typology one that prioritizes important key factors such as carbon performance, biodiversity, and urban integration over mere technical utility developed in collaboration with:

## Why Change?

There are many problems but to make it easier, I would like to give some examples with my case study building. There is no apparent flow of logistics at the location, leading to operational malfunctioning. The absence of efficient traffic routes means that vehicles have to maneuver awkwardly

Entrance inefficiency makes worse, since trucks have to queue in the road network to enter the premises, delaying the entire process and undermining the efficiency of the supply chain.

Urban visualization suggests that the location serves as an eyesore that adversely affects the quality of life for residents around the area.

Moreover, the lack of communal spaces shows that the project is not only environmentally unfriendly but also socially irresponsible, lacking any elements that cater to workers' well-being. This results in the creation of a "dead zone" within the urban environment.

This area constitutes an environmental risk because it has no sustainable environmental features whatsoever.

The industrial surfaces are expansive and there is very little ecological interaction within the facility. This implies that a high heat island effect exists, alongside the problem of uncontrolled stormwater drainage.

No evidence of sustainable systems

Pollution

No traffic circulation

Poor Entrance Efficiency

Minimal ecological integration.

Visual Blight on Residential Neighbors

No Common Spaces



Case Study

The concept is to transform the Logistics Center into a “Community Integrated Logistics”. This constitutes a paradigm shift in industrial planning, setting a new benchmark in which the infrastructure acts as a Neighborhood Resource.

The use of a unique architecture that adopts terraces at ground level allows the development to protect the views from neighboring homes

The basis of this design is founded upon the principles of modal segregation and efficient traffic flow. The isolation of industrial uses from the local streets creates zero neighborhood congestion.

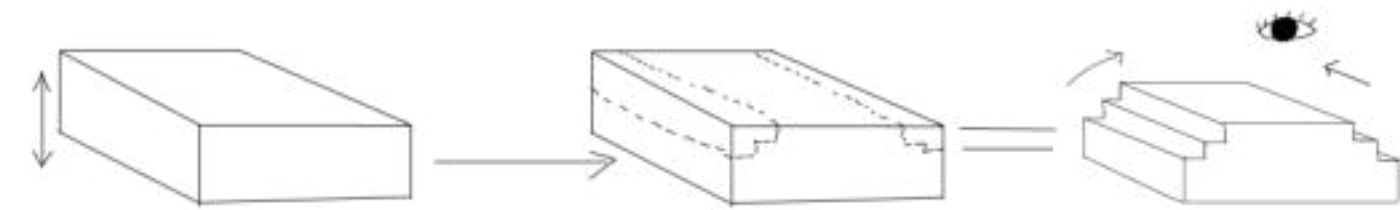
This is made possible through dense vegetation and tree canopy streets which act as a green filter system using stormwater management techniques to counteract issues of noise pollution and heat gain.

In the first approach, i am introducing terraced architecture at a low level that reduces the mass of the building.

In essence, this geometrical change aims to protect the views of the residents since it guarantees their line of sight to the horizon.

By offsetting the massing of the structure from the edges, the design effectively protects the views for residences while avoiding the negative impact of a “grey wall” which is common in industrial districts.

Such an approach also results in large-scale horizontal planes that are used for intensive vegetation coverage, functioning as a natural sound barrier to dampen noise pollution, Overall, this approach transforms a proprietary service into a community service.

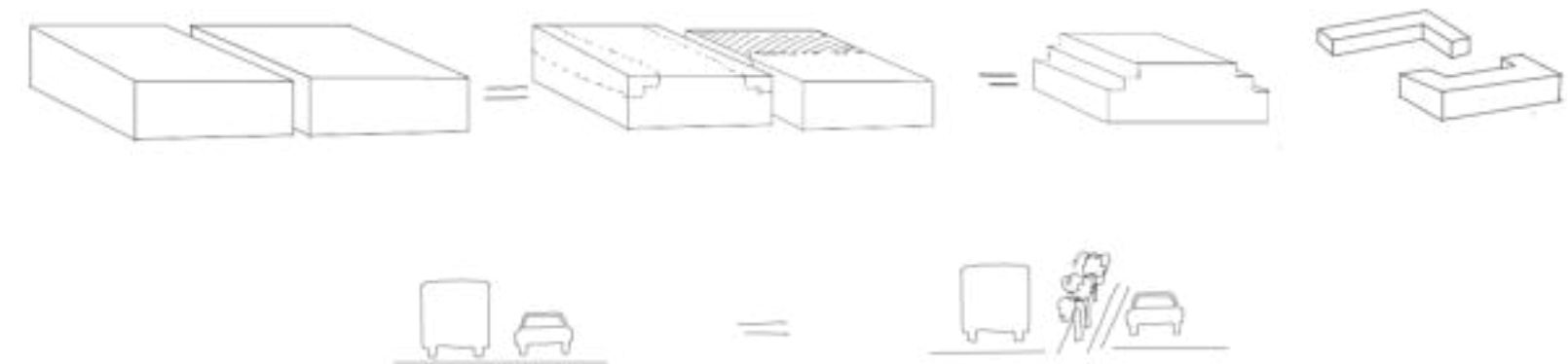


This second approach is creating a physical division between storage area and office, control areas to avoid any interaction in the process.

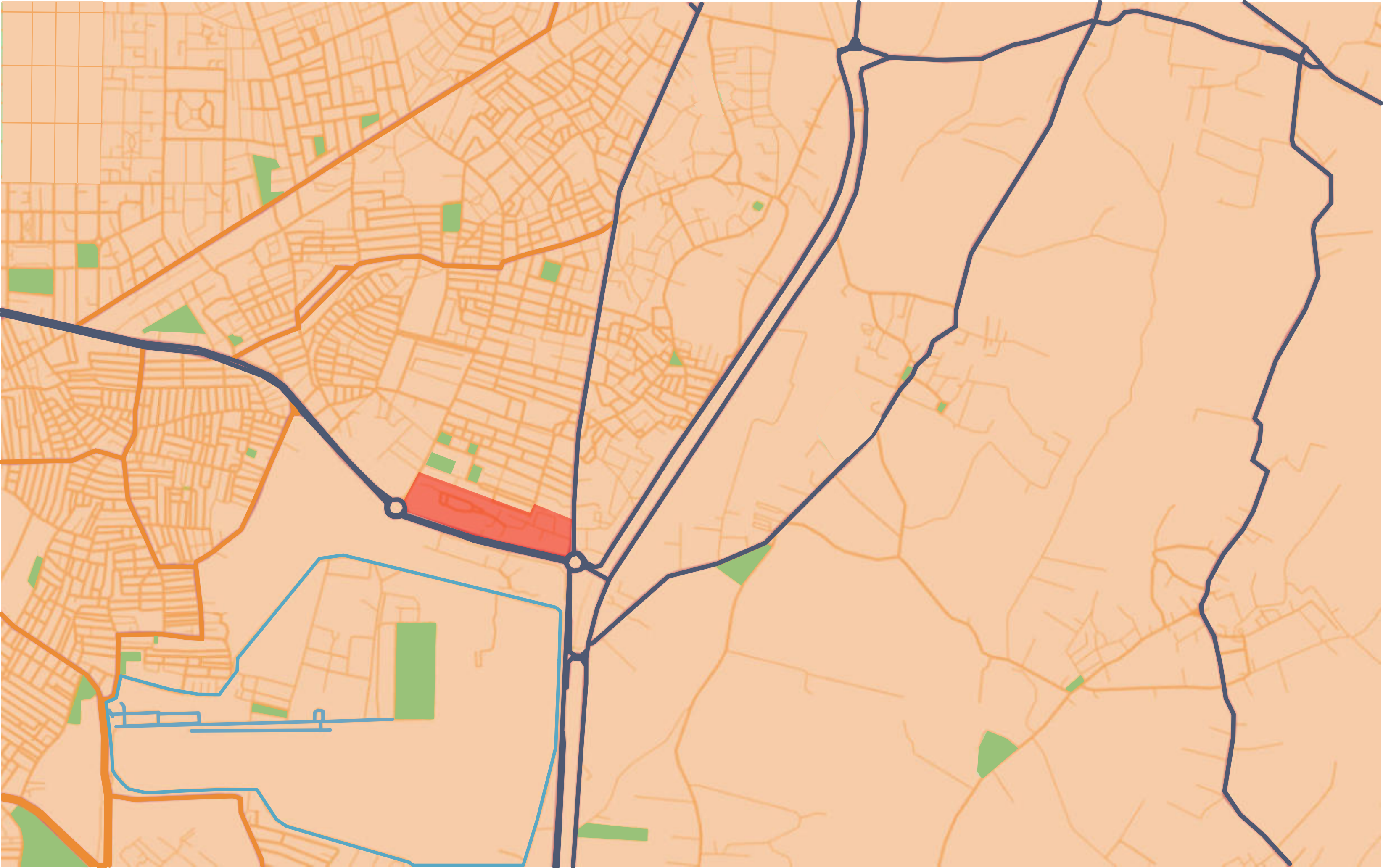
In addition, the design of the site layout creates an effective separation between Trucks and private car traffic. Through the application of tree-canopy streets and bioswales within this organized system of paths, the infrastructure is kept private and operates without any interaction with the nearby neighborhood street network.

The segregation of functions and traffic in the building will avoid all the usual problems that arise from the lack of separation and cause noise and congestion.

With this approach, it will be possible to keep different purposes areas away from eachother. One of the major advantages of this design is that the project will leave a much smaller ecological impact on its surroundings because of landscape engineering and bioswale technology that naturally clean the water and air in the area through natural filtration systems.







The site location is Yunuseli, Osmangazi region of Bursa, Türkiye, that holds immense potential due to its geographical location that connects important industrial zones with expansion zones of the residential sector.

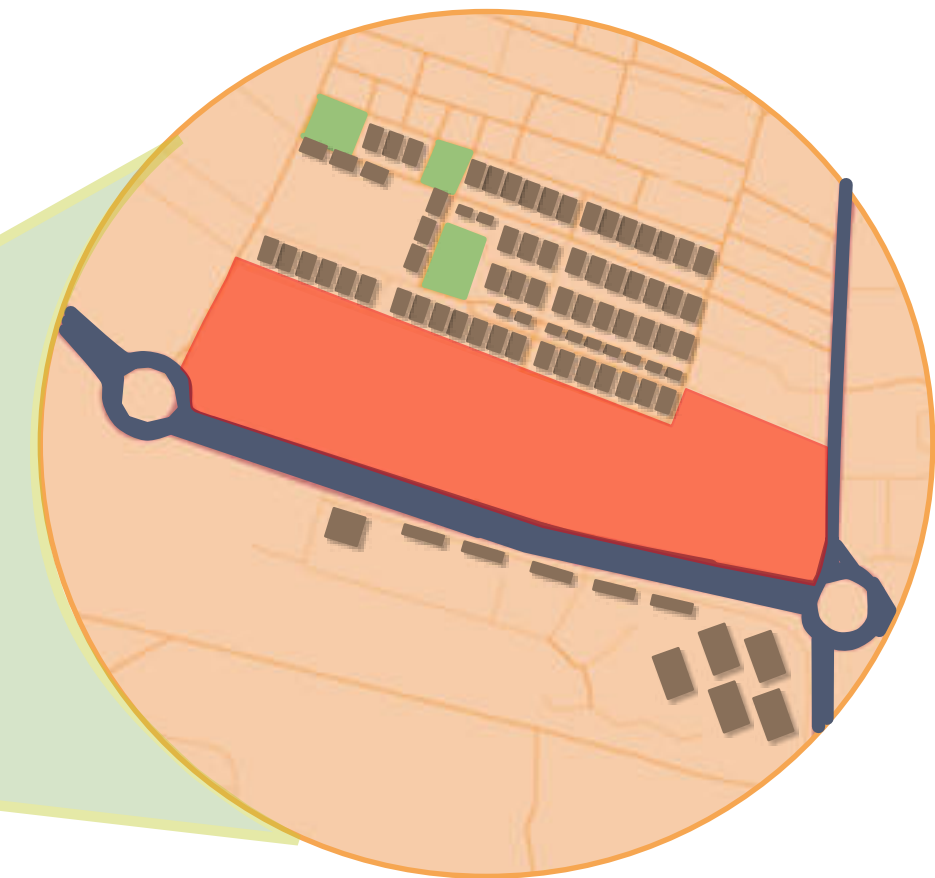
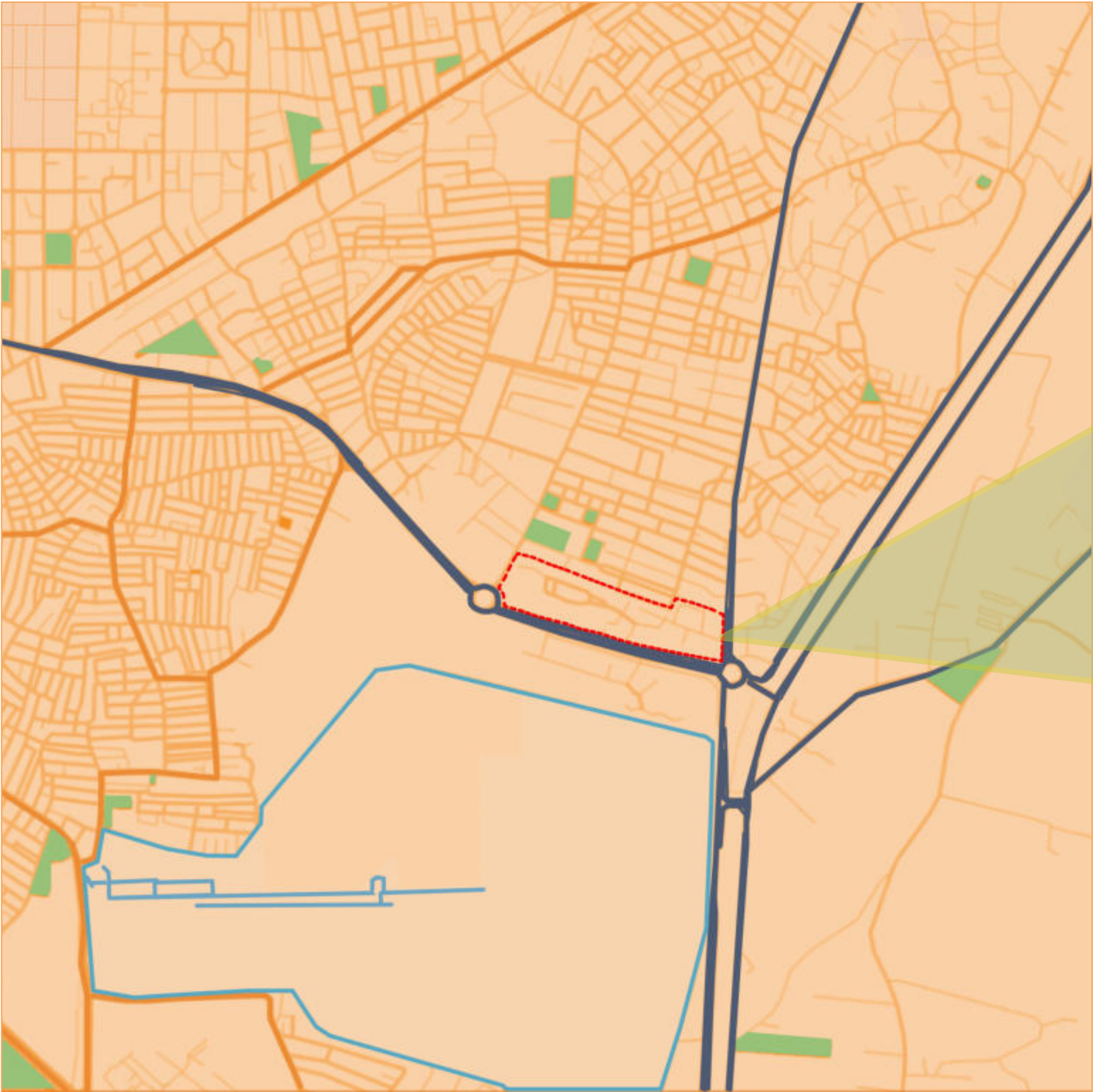
The historical significance of Yunuseli lies in the fact that it was the birthplace of the first airstrip of Bursa the Yunuseli Airport.

Osmangazi's reputation as an industrial hub, home to some of the most important industrial zones like automotive and steel industries, makes it an advantage for the site. Yunuseli is a special combination of demographics that includes mainly affluent and industrialists, who require a more refined level of urban quality.

This place is characterized by a shift from conventional industrial storage areas to high end logistics facilities within the context of Bursa's "Green City" concept, where agriculture meets industrialization.

Bursa's progression towards becoming the major logistics center of Turkey leaves Yunuseli at the junction of logistics and social integration. The development of the area is based on the requirement of balancing the industrial heritage of the place with sustainability requirements. Any future development plans for the area include overcoming the issue of "unplanned urbanization" in previous areas through improved stormwater systems and modal separation, thereby making logistics facilities an integral part of the Yunuseli community.

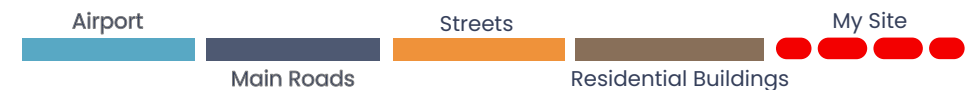




The location of this site is strategically positioned in the center of an important transportation hub within Yunuseli. It lies directly near the network of the main roads. These roads form the backbone of the regional connections, combining the site to the greater Bursa transportation network. Near the site lies the Yunuseli Airport. This makes the site a highly accessible industrial location.

One of the main aspects that define the character of the immediate context of the site is its connection to the adjacent residential area.

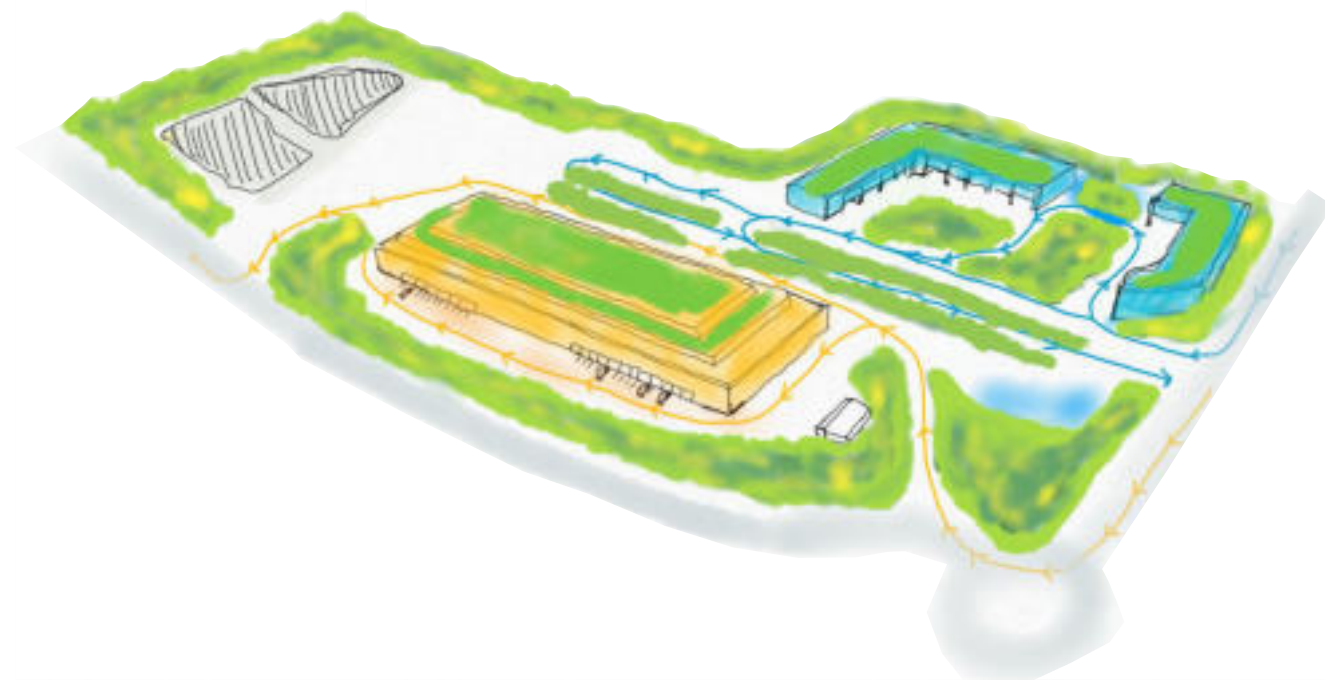
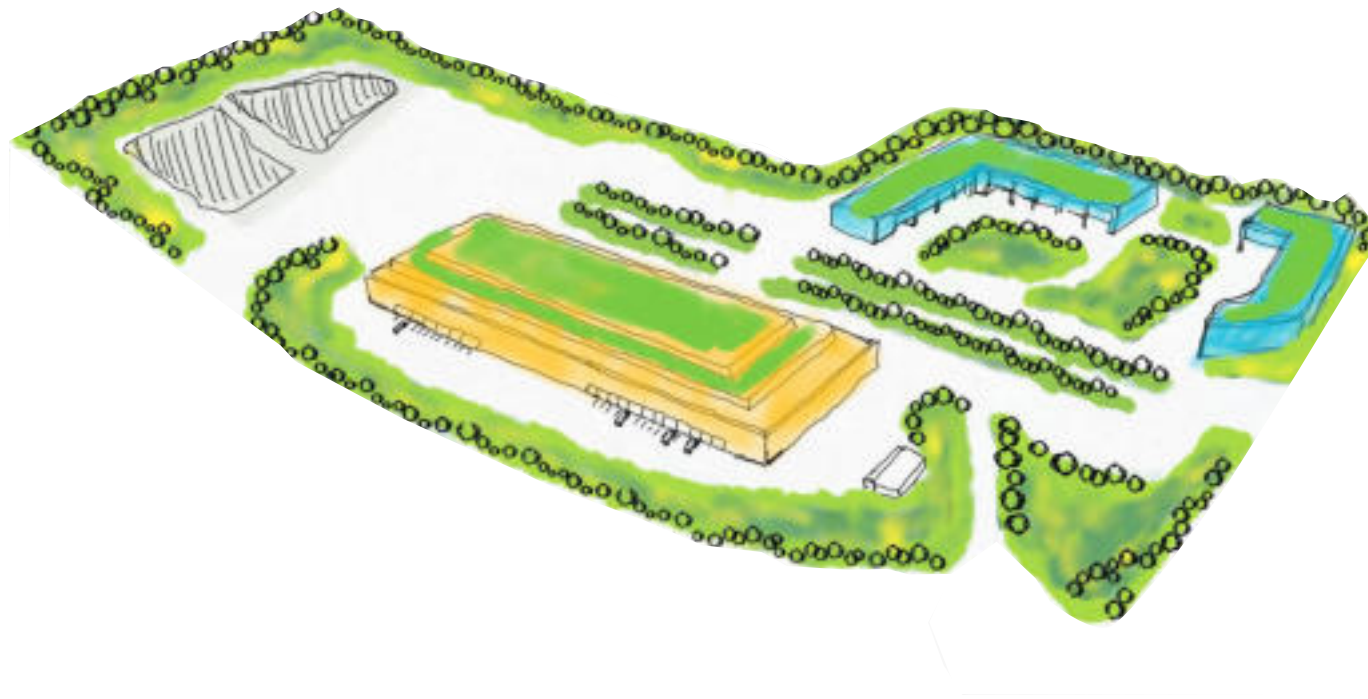
In the close up view, the site (red) is surrounded by several residential homes, thus presenting an interesting urban phenomenon where industrial activities and the residential sector are connected through direct physical borders.



# SKETCH

This sketch emphasize the importance of sustainability in the project and its function as an "eco systemic filter." With the site being enclosed by dense vegetation and a consistent canopy of trees, it is possible to create a natural barrier against pollution and noise from industries.

It is clear from the diagram how the use of bioswales and landscape engineering works.



The second sketch is indicative of the base for the operations carried out within the site through planning regarding traffic flow. It shows that primary and secondary transport are segregated to ensure that there are no disruptions within the entrance process. The design ensures that the activities taking place inside do not encroach into the urban areas surrounding the



# DESIGN

Problems

Solutions

|                                              |                   |                                         |
|----------------------------------------------|-------------------|-----------------------------------------|
| Residential Visual Preservation Index (RVPI) | %                 | The buildings block the neighbors view. |
| Internal traffic circulation                 | vehicles/hour     | Only one entrance, 0 efficiency.        |
| Environmental Performance Improvement        | %                 | There is no sustainable solutions.      |
| Biodiversity                                 | species count, m² | None, there is no suitable place.       |
| Vegetation                                   | green area (m²)   | Lack of greenary                        |

|                                                |                   |                                                                              |
|------------------------------------------------|-------------------|------------------------------------------------------------------------------|
| Residential Visual Preservation Index (RVPI) % | %                 | Terrace roof                                                                 |
| Internal traffic circulation                   | vehicles/hour     | one additional entrance and new traffic system on the site                   |
| Environmental Performance Improvement          | %                 | stormwater systems, bioswales and green roof                                 |
| Biodiversity                                   | species count, m² | with varient Vegetation on site, different animals can live and move freely. |
| Vegetation                                     | green area (m²)   | greenary all around the site                                                 |

# Alternative Design 1



Storage/Cross-Dock



Offices/ Control Points

In this case, i tried to work with two huge buildings.

Though the arrangement had clarity for visual movement, it had an adverse effect by causing what can be called a “tunnel effect” that was more industrial than welcoming. Not only did this fail in considering the natural flow of the site, but it was also an unfriendly “wall” that the resident neighbors had to put up with.

This could have been avoided with the change in shape to form the U and L shaped buildings which I achived with my design

|                                              |   |
|----------------------------------------------|---|
| Residential Visual Preservation Index (RVPI) | % |
| Environmental Performance Improvement        | % |

# Alternative Design 2



In this design, i tried out a less centralized design where the buildings were separated into individual blocks, such as the C-shaped block and the separate blocks.

Although there was some element of diversity here, it did not result in either a coherent identity for the campus or an effective flow of operations.

The vast distance between each mass produced "dead zones" within the landscape which lacked any real use, while the absence of an axis made navigating around the site a difficult task for all.

|                              |               |
|------------------------------|---------------|
| Internal traffic circulation | vehicles/hour |
|------------------------------|---------------|

A Life Cycle Assessment (LCA) is an elaborate process that aims to assess the environmental effects of a structure or a product across its entire life cycle. The process is referred to as “From Cradle to Grave” which focuses on the measurement of resource use and emissions with a scientific perspective toward sustainability

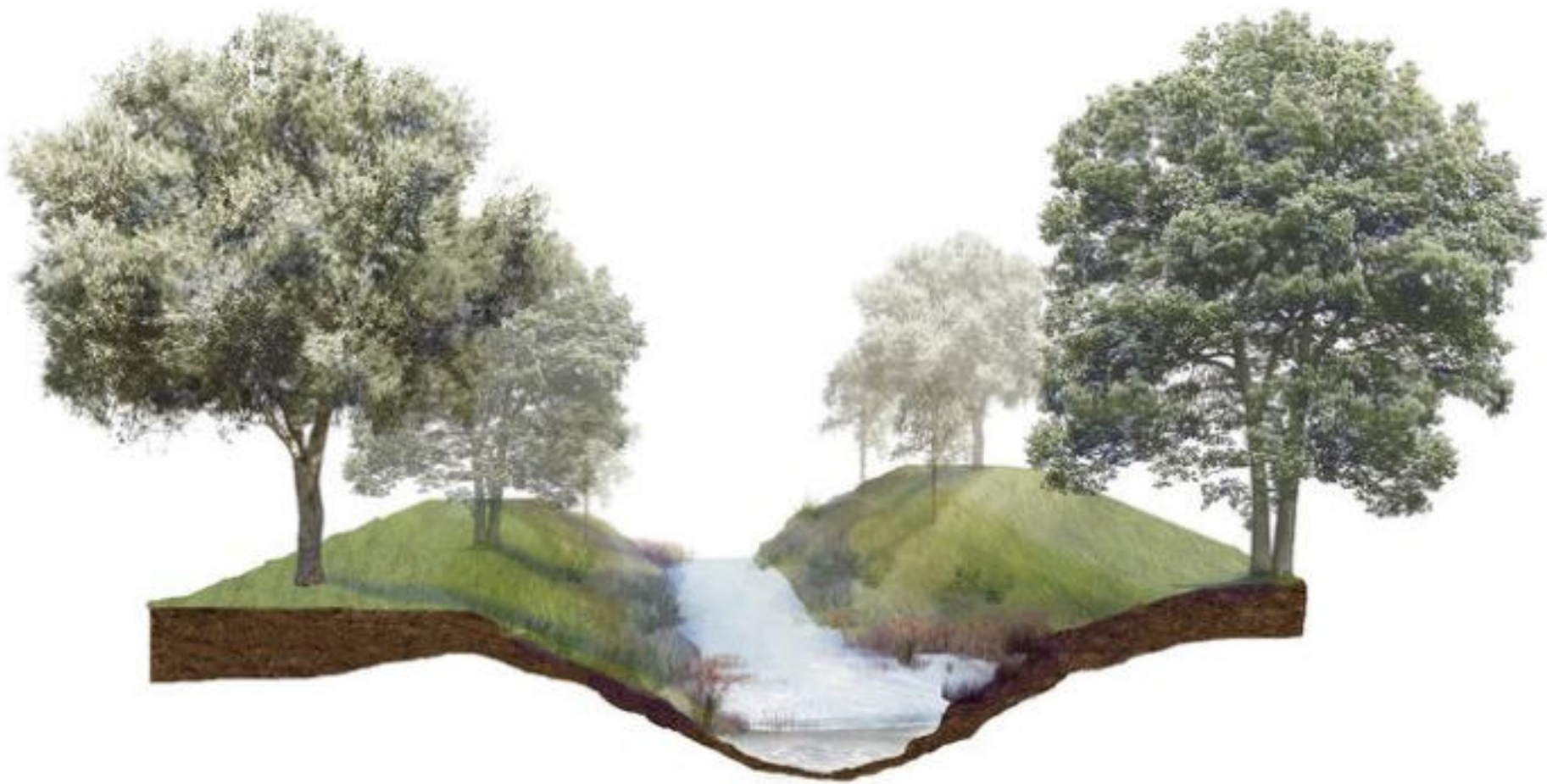
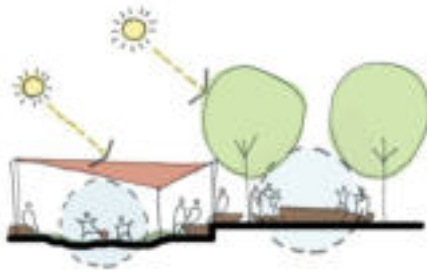
The Resources phase is when decisions are made about using timber that does not have much carbon, using recycled steel, or using paving systems that allow for water permeation. In choosing materials with low embodied energy, the environmental impact is minimized from the onset.

The process of sustainability persists after the life span of operation of the structure. In LCA, the “End of Life phase involves disassembly and recycling. The present project will facilitate the dismantling process so that the components may be utilized again.

Logistics is an essential component of the LCA process, which contributes a substantial share of greenhouse gas emissions. It can be seen from the diagram that "Distribution" acts as a bridge between production and use. Thanks to effective traffic management and modal segregation, our design reduces the amount of emissions from heavy transport during idling and fuel consumption.



There is also a software! One Click LCA is an international standard software for conducting life cycle assessments on buildings that allows architects and engineers to compute the environmental impacts of their structures in a very accurate manner. It accomplishes this by making the process of collecting information from BIM models and spreadsheets automated and therefore avoids the need for any additional work on the part of users. This software forms the crucial link between the designer's vision and environmental reality by allowing users to assess GWP and other environmental performance criteria of a building through its whole life cycle.



The first benefit associated with this platform is the availability of a comprehensive and validated database of Environmental Product Declarations. This feature will enable the user to conduct comparative assessments of the carbon content of different structural systems such as timber, steel, and concrete at an early stage of designing. The advantage of using localized data for comparison will allow the architect to make a sustainable choice, which will have zero or low embodied carbon content..

For highlighting such results in a project, one must concentrate on the story of "Carbon Payback." A professional would be able to make use of the software to prove that some particular critical decision-making steps such as selecting cross-laminated timber structure or high-performance insulation are responsible for bringing down the tons of \$CO\_2\$ equivalent. With the help of comparative bar charts and heat maps, one would be able to make people understand the hidden costs of constructing buildings.

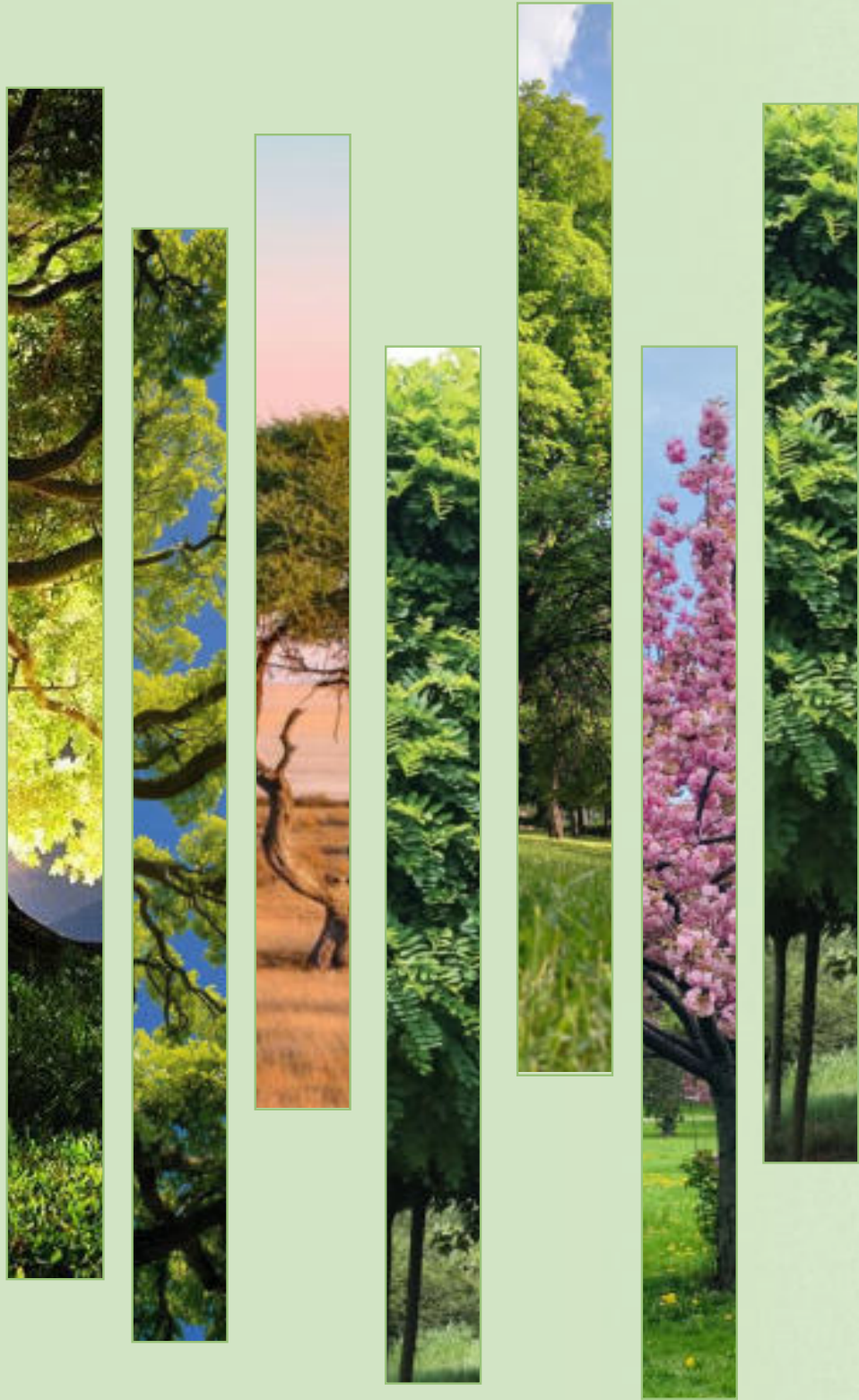
Furthermore, integrating One Click LCA into a project workflow facilitates seamless compliance with global green building certifications such as LEED, BREEAM, and DGNB. Beyond carbon, the tool allows users to assess circularity and life-cycle costing, providing a holistic view of a building's value over time.

# Biodiversity

The Yunuseli district and Bursa as a city has specific mix of native and adaptive flora: Maple, Oriental Plane, Persian Silk Tree, Linden , and Acacia.

The planting of such trees as the Oriental Plane and Linden, as stated above, was deliberately done in order to retain the environmental genetics of the neighborhood. The trees mentioned have survived well in the harsh Bursa weather conditions as well as the Yunuseli soil.

The high density of the plantings, which is replicated from the existing biodiversity, makes the project even more well planed. In doing so, the planting will be a real reflection of nature as opposed to being a constructed feature.



With the mapping of the varied wildlife, the existing wildlife in the Yunuseli area is revealed to include such species as the indigenous birds, hedgehogs, hares, and other pollinators. In order to avoid any displacement of these wildlife species, the design of the facility includes trees and shrubs on which they depend.

The application of high density foliage as well as multi-layered canopy is a conscious choice made with the specific intention of catering to the demands of current inhabitants in the area, whether they are animal or human. In addition to specific bushes and low-level plants designed to serve as safe places for small mammals, trees like the Oriental Plane Tree and the Silk Trees will act as refuges for birds within the area. The creation of such an environment based on the needs of the animals present at the site will reduce the negative effects that industrialization may have had on wildlife in the area.



END